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Research Statement

My primary research fields are empirical asset pricing and investment, focusing on risk pricing, financial market frictions, and the design of investment strategies. I also study how fiscal policy interacts with macroeconomic conditions and financial markets, analyzing the mechanisms through which fiscal shocks drive inflation and affect asset prices. Methodologically, my work employs advanced empirical methods and large-scale data analysis to address interesting questions in asset pricing and macro-finance.

Empirical Asset Pricing and Investment

My current research investigates zero-beta rates and zero-beta (factor-neutral) investment strategies. In my job market paper, **“Resolving the Zero-Beta Rate Puzzle”**, I show that the persistently high estimated zero-beta rates across asset pricing models arise from pervasive factor model misspecification rather than a genuinely high risk-free rate or convenience yield. When a factor model does not perfectly price assets, the implied zero-beta rate becomes non-unique and tends to be upward biased. I introduce a new investment-based measure of misspecification—the maximum Sharpe ratio attainable by zero-investment, zero-beta portfolios—which quantifies the economic magnitude of pricing errors. Using both classical and machine-learning factor models, I find substantial and systematic misspecification, explaining their uniformly elevated zero-beta rates. Simulations confirm that realistic levels of misspecification can fully reproduce the empirical puzzle even when the true risk-free rate is low. Furthermore, misspecification implies profitable zero-beta investment strategies that remain robust after accounting for realistic transaction costs. Together, these results reinterpret the zero-beta rate puzzle as a statistical artifact of imperfect factor pricing, offering new insights on the risk-free rate, risk premium, and convenience yield.

In a related paper, **“Zero-Beta (Factor-Neutral) Portfolios”**, I propose a unified framework, based on the maximum Sharpe ratio attainable by zero-investment zero-beta portfolios, which facilitates consistent out-of-sample testing and comparison of factor models while accounting

for transaction costs. The framework applies broadly to tradable and non-tradable factors, nested and non-nested specifications, and both traditional and modern machine-learning-based conditional models. The analysis shows that conventional model comparisons based on the Sharpe ratios of factor portfolios can be misleading: once transaction costs are considered, the after-cost Sharpe ratios of factors generated by machine-learning models often become significantly negative, understating their true pricing performance. Finally, the results highlight an important investment implication—systematically trading model mispricings through beta-neutral (zero-beta) strategies proves more profitable and implementable than harvesting factor risk premia directly, achieving superior net-of-cost performance.

Building on my previous findings that even state-of-the-art factor models—including those based on advanced machine-learning techniques—remain substantially misspecified, I explore the sources of these misspecifications in a new project, **“What Has Gone Wrong in the Best Factor Models”**. In this paper, I use zero-beta portfolios as diagnostic tools to trace where risk pricing breaks down. By analyzing the behavior of zero-beta returns across model specifications, asset groups, and market conditions, I aim to identify whether the underlying issues stem from missing risk factors, model instability, non-linear pricing, or limits to arbitrage. The goal is to move beyond documenting the existence of misspecification toward understanding its economic origins, offering new insights into why modern factor models, despite their sophistication, continue to leave large and systematic pricing errors unexplained.

Fiscal Policy and Fiscal Shocks

A second branch of my research studies the interaction between fiscal policy, macroeconomics, and financial markets. In **“Fiscal Policy and the Government Debt Maturity Structure”**, I use vector autoregressions (VAR) with external instruments to identify fiscal shocks and examine how their macroeconomic effects depend on the government debt maturity structure using local projections. I find that longer debt maturity significantly attenuates the output and inflation responses to fiscal expansions. Using a structural Fiscal Theory of the Price Level (FTPL) model, I show that longer maturity enables the government to realize capital gains on outstanding long-term debt after a deficit shock, reducing the incentive to inflate away existing liabilities. This mechanism explains the state-dependent transmission of fiscal policy observed in the data.

In a new and ongoing project, **“Fiscal Shocks, Beliefs, and Asset Prices”** (with [Anna Cieslak](#) and [Hao Pang](#)), we develop a narrative, high-frequency identification of fiscal shocks using Bloomberg hot headlines data to examine how fiscal news shapes investor beliefs and financial market reactions. We perform large-scale textual analysis on over one million Wall Street Journal articles from ProQuest’s TDM Studio to quantify fiscal sentiment and belief dynamics, and study their influence on asset prices. Applying the Cieslak–Pang decomposition ([Cieslak and Pang, 2021](#)), we further separate fiscal shocks into growth, inflation, and risk premium components, allowing us to identify which dimensions of beliefs drive asset price responses.